Course Name: Database Management Systems 428768

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modelling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyse the existing design of a database schema and apply concepts of normalization to design an optimal database and implement transaction management
CO7: Discuss concurrency & indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: **University Roll No. .....**

End Term Examination, Odd Semester 2025-26

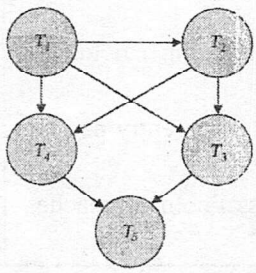
B.Tech (CSE), Year: II, Semester: III B.Tech.(ME) IIISem

BCSC 1003, Database Management Systems

Time: 3 Hours **Maximum Marks: 45**

Section – A

Attempt All Questions 3 X 6 = 18 Marks

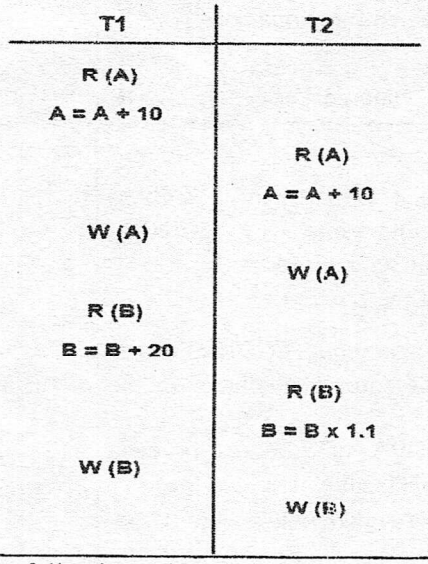
No.	Detail of Question	Marks	CO	BL	KL
1	Consider the precedence graph of Figure 1. Is the corresponding schedule conflict serializable? Explain your answer.  Figure 1. Precedence graph	3	6	U	C
2	During its execution, a transaction passes through several states, until it finally commits or aborts. List all possible sequences of states through which a transaction may pass. Explain why each state transition may occur.	3	6	R	F
3	Consider a relation R (A, B, C, D, E) with the following set of Functional Dependencies (FDs): 1. A → BC 2. C → D 3. D → E	3	6	A	P

	The relation is decomposed into the following sub-relations: R1(A, B, C), R2(C, D, E) & R3(A, D) Determine whether the decomposition {R1, R2, R3} is lossless or lossy.				
4	What is a recoverable schedule? Why is recoverability of schedules desirable? Are there any circumstances under which it would be desirable to allow non-recoverable schedules? Explain your answer.	3	8	U	C
5	Explain the Timestamp-Based Concurrency Control Protocol in database systems. In your answer, mention: a) How timestamps are used to maintain serializability. b) What happens when a transaction violates the timestamp order.	3	7	R	F
6	A university maintains a Student database file containing 10,000 student records, each of fixed length. Each disk block can store 100 records (blocking factor = 100). To improve search performance on the Roll Number (primary key) field, the database administrator considers creating an index file. Answer the following: a) If a dense index is created on the Roll Number field, how many index entries will exist? b) If a sparse index is created with one index entry per data block, how many index entries will exist? c) Explain the performance implications of both approaches for searching and maintenance.	3	7	A	P

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
7	Consider a relation R(Employee, Project, Skill) with the following facts: • An employee can work on multiple projects. • An employee can have multiple skills. • Projects and skills are independent of each other for the same employee. a) What is a Multivalued Dependency (MVD)? Explain its	5	6	An	P

	significance in database normalization. b) Identify all MVDs that exist in the given relation. c) State whether the relation R is in Fourth Normal Form (4NF) and justify your answer.				
8	Check whether the given schedule S is view serializable or not- 	5	6	An	P
9	Consider the following schedule involving two transactions T1 and T2 on data items X and Y: Step Operation 1 T1: Lock-X(X) 2 T1: Read(X) 3 T2: Lock-X(Y) 4 T2: Read(Y) 5 T1: Lock-X(Y) 6 T2: Lock-X(X) 7 T1: Write(Y) 8 T2: Write(X) a) Identify whether the given schedule follows the Two-Phase Locking (2PL) protocol. b) Determine if deadlock occurs in this schedule. c) If a deadlock occurs, show how it can be detected and resolved.	5	7	A	P

Section – C

*Attempt **All** Questions* 4 X 3 = 12 Marks

No.	Detail of Question	Marks	CO	BL	KL
10	For relation R(A, B, C, D, E) the dependencies are: $F = \{A \rightarrow BC, B \rightarrow D, CD \rightarrow E, A \rightarrow D, E \rightarrow A\}$ a) Identify and explicitly state which functional dependencies in F are redundant. b) Show all steps to compute the canonical cover (F_c) for R.	4	2	A	P
11	Given Database Schema: STUDENT(StudentID, Name, Department, CGPA) COURSE(CourseID, CourseName, Credits) ENROLLMENT(StudentID, CourseID, Grade) Write an SQL query for the following: a) To list the names of all students in the STUDENT table. b) To find the total number of courses available in the COURSE table. c) To display the average CGPA of students in each department from the STUDENT table. d) To display the student's name, course name, and grade for students who scored grade 'A'.	4	4	A	P
12	A company maintains a centralized database with the following relation: EMPLOYEE (EmpID, Name, Department, Location, Salary, Experience) To improve performance and availability, the company decides to distribute the database across its Delhi, Mumbai, and Bengaluru branches. a) Propose a suitable horizontal fragmentation of the EMPLOYEE relation so that each branch stores data relevant to its own employees. b) Suggest a vertical fragmentation that separates frequently accessed employee information by the HR Department and Payroll Department. c) Explain one advantage and one disadvantage of using both horizontal and vertical fragmentation together in a distributed database.	4	8	A	C